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1 Collaboration Opportunities

1.1 Potential Partners for Research, Funding, and Deployment

Academic, Industry, and Community Partnerships

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1.2 Table of Contents

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1.3 1. Overview

Successful development and deployment of AI-based IDS for next-generation networks requires collaboration across academia, industry, standards organizations, and open-source communities. This document identifies potential partners and outlines mutually beneficial collaboration opportunities.

1.3.1 1.1 Collaboration Objectives

- **Data Access:** Obtain real-world network traffic datasets
 - **Equipment Donation:** Secure testbed hardware (SDN switches, IoT devices)
 - **Financial Support:** Research funding, fellowships, cloud credits
 - **Validation:** Deploy IDS in production networks for real-world testing
 - **Knowledge Exchange:** Access to domain experts, joint publications
 - **Technology Transfer:** Industry adoption of research outcomes
-

1.4 2. Academic Institutions

1.4.1 2.1 Leading Universities in Network Security

Carnegie Mellon University (CMU) - **Department:** CyLab Security and Privacy Institute - **Expertise:** Network security, ML for security - **Potential Collaboration:** - Joint research projects - Student exchange programs - Dataset sharing - Joint publications - **Contact:** Faculty in Computer Science Department

University of California, Berkeley - Department: International Computer Science Institute (ICSI) - **Expertise:** Network measurement, security, machine learning - **Potential Collaboration:** - Access to network traces - Collaboration with NetSys Lab - Joint workshops - **Contact:** Prof. Vern Paxson (security expert)

MIT - Department: Computer Science and Artificial Intelligence Laboratory (CSAIL) - **Expertise:** AI, cybersecurity, networks - **Potential Collaboration:** - AI/ML methodology guidance - Research seminars and visits - Joint publications

Georgia Tech - Department: Institute for Information Security & Privacy - **Expertise:** Network security, SDN, IoT security - **Potential Collaboration:** - SDN testbed collaboration - Joint experiments - Faculty mentorship

University of Illinois Urbana-Champaign - Department: Coordinated Science Laboratory - **Expertise:** Network security, distributed systems - **Potential Collaboration:** - Federated learning research - Distributed IDS deployment

1.4.2 2.2 International Collaborations

UNSW Sydney (Australia) - Expertise: IDS datasets (UNSW-NB15), security research - **Collaboration:** Dataset usage, joint evaluation

Technical University of Munich (Germany) - Expertise: 5G/6G security, network slicing - **Collaboration:** 5G testbed experiments, joint papers

National University of Singapore (NUS) - Expertise: AI, IoT security - **Collaboration:** IoT malware research, federated learning

1.5 3. Industry Partners

1.5.1 3.1 Telecommunications Operators

Verizon - Interest: 5G network security, edge computing - **Potential Collaboration:** - Access to anonymized 5G traffic data - Testbed deployment in real 5G network - Joint research on 5G-specific attacks - **Benefits to Verizon:** - Advanced threat detection for 5G core - Protection of network slicing infrastructure - Reduced operational security costs - **Contact:** Verizon Innovation Program, CTO Office

AT&T - Interest: Network security, SDN/NFV protection - **Potential Collaboration:** - AT&T Research collaboration - Dataset provision - Proof-of-concept deployment - **Benefits to AT&T:** - Enhanced security for virtualized networks - AI-driven automation for threat response - Competitive advantage in secure 5G - **Contact:** AT&T Labs Research

Deutsche Telekom - Interest: European 5G security, GDPR compliance - **Potential Collaboration:** - Privacy-preserving IDS validation - Federated learning deployment across regions - **Benefits:** - GDPR-compliant threat detection - Cross-border threat intelligence - **Contact:** Deutsche Telekom Innovation Labs

T-Mobile - Interest: IoT connectivity, mobile network security - **Potential Collaboration:** - IoT traffic analysis - Edge-based IDS for massive IoT - **Benefits:** - Protection for IoT platform - Scalable security for millions of devices

1.5.2 3.2 Network Security Vendors

Cisco Systems - Interest: Next-gen firewalls, SD-WAN security, Cisco Umbrella - **Potential Collaboration:** - Equipment donation (switches, routers) - Joint development of AI-powered Cisco IDS - Cisco Research grant - Integration with Cisco Security products - **Benefits to Cisco:** - AI/ML enhancements to product portfolio - Academic validation of technology - Early access to cutting-edge research - **Contact:** Cisco Research, DevNet

Palo Alto Networks - Interest: ML-powered threat prevention, cloud security - **Potential Collaboration:** - Integration with Cortex XDR - Joint research on adversarial ML - Funding for explainable AI research - **Benefits:** - Enhanced AI capabilities in products - Explainability for regulatory compliance - **Contact:** Palo Alto Networks Research

Fortinet - Interest: SD-WAN security, FortiGuard Threat Intelligence - **Potential Collaboration:** - Threat intelligence data sharing - Joint threat hunting research - Equipment provision (FortiGate devices) - **Benefits:** - AI-enhanced threat detection - Improved threat intelligence accuracy

Darktrace - Interest: AI-driven autonomous response, self-learning IDS - **Potential Collaboration:** - Reinforcement learning for threat response - Benchmark comparisons - Sponsored research - **Benefits:** - Academic validation of AI approach - Enhancements to self-learning algorithms

1.5.3 3.3 Cloud and Technology Giants

Amazon Web Services (AWS) - Interest: Cloud security, AWS Shield, GuardDuty - **Potential Collaboration:** - AWS Research Credits (\$10-20K) - AWS Security Heroes program - Integration with AWS security services - **Benefits to AWS:** - Enhanced threat detection for VPC - AI models for GuardDuty - **Contact:** AWS Research, AWS Security team

Google Cloud - Interest: GCP security, Chronicle Security - **Potential Collaboration:** - Google Cloud Research Credits - Joint research with Chronicle team - TensorFlow optimization for IDS - **Benefits:** - Improved security for Google Cloud customers - TensorFlow use case showcase

Microsoft - Interest: Azure Security, Microsoft Defender - **Potential Collaboration:** - Microsoft Research Fellowship - Azure credits for research - Integration with Azure Sentinel - **Benefits:** - AI/ML enhancements to Defender - Azure security product improvement - **Contact:** Microsoft Research, Azure Security team

IBM - Interest: QRadar, Watson for Cyber Security - **Potential Collaboration:** - IBM Research collaboration - Quantum-safe security research - Funding for AI security research - **Benefits:** - Watson AI enhancements - Integration with QRadar SIEM

1.5.4 3.4 IoT and Edge Computing

NVIDIA - Interest: Jetson edge AI, security for autonomous systems - **Potential Collaboration:** - Jetson device donation for edge IDS - NVIDIA GPU Cloud credits - Joint optimization for embedded inference - **Benefits:** - Showcase for Jetson AI capabilities - Security for NVIDIA edge platforms - **Contact:** NVIDIA Research, Jetson Developer Program

Arm - Interest: IoT security, TrustZone, secure edge computing - **Potential Collaboration:** - Arm Research grant - Optimization for Arm Cortex-M/Cortex-A - Integration with Arm Mbed OS security - **Benefits:** - AI security for Arm-based IoT devices - Efficient inference on Arm processors

Qualcomm - Interest: 5G chipsets, edge AI acceleration - **Potential Collaboration:** - Access to 5G test equipment - On-device AI optimization - **Benefits:** - Security for Qualcomm 5G modems - AI acceleration for Snapdragon

1.6 4. Research Laboratories

1.6.1 4.1 Industrial Research Labs

Bell Labs (Nokia) - Expertise: 5G/6G, network security, AI - **Potential Collaboration:** - Joint research on 6G security - Access to experimental 5G networks - Visiting researcher program - **Benefits:** - Early insights into 6G security challenges - Real-world 5G deployment validation

IBM Research - Expertise: AI, quantum computing, security - **Potential Collaboration:** - IBM Research Fellowship - Quantum-resistant IDS research - Joint publications - **Benefits:** - AI/quantum expertise - High-impact research collaboration

Microsoft Research - Expertise: ML, systems, security - **Potential Collaboration:** - Research internship - Joint projects on adversarial ML - Access to large-scale datasets - **Benefits:** - Leading-edge AI research - Industry perspective on deployability

Google Research - Expertise: Deep learning, federated learning, privacy - **Potential Collaboration:** - Federated learning research - Privacy-preserving ML - Intern/visiting scholar programs - **Benefits:** - Expertise in FL and differential privacy - TensorFlow optimization guidance

1.6.2 4.2 Government and National Labs

NSF (National Science Foundation) - Programs: CNS (Computer and Network Systems), SaTC (Secure and Trustworthy Cyberspace) - **Potential Collaboration:** - NSF Graduate Research Fellowship (GRFP) - Research grants (CAREER, SaTC) - **Benefits:** - Significant research funding (\$500K-1M for CAREER)

DARPA (Defense Advanced Research Projects Agency) - Programs: AI Next, Assured Autonomy - **Potential Collaboration:** - DARPA Young Faculty Award - Collaborative research programs - **Benefits:** - High-risk, high-reward research funding - National security impact

NIST (National Institute of Standards and Technology) - **Interest:** Cybersecurity standards, AI standards - **Potential Collaboration:** - AI Risk Management Framework input - Evaluation of IDS against NIST standards - **Benefits:** - Standards alignment - Government credibility

1.7 5. Standardization Bodies

1.7.1 5.1 IETF (Internet Engineering Task Force)

- **Working Groups:** DOTS (DDoS Open Threat Signaling), I2NSF (Interface to Network Security Functions)
- **Potential Collaboration:**
 - Propose draft RFC for AI-based IDS interfaces
 - Participate in working groups
 - Standardize federated threat intelligence sharing
- **Benefits:**
 - Industry adoption through standardization
 - Shape future IDS protocols

1.7.2 5.2 3GPP (3rd Generation Partnership Project)

- **Focus:** 5G/6G security specifications
- **Potential Collaboration:**
 - Contribute to 5G security standards (TS 33.xxx series)
 - Propose AI-based security functions for 6G
- **Benefits:**
 - Influence 5G/6G security architecture
 - Early access to specifications

1.7.3 5.3 ETSI (European Telecommunications Standards Institute)

- **Groups:** NFV Security, Cyber Security Technical Committee
- **Potential Collaboration:**
 - NFV security standardization
 - AI in cybersecurity best practices
- **Benefits:**
 - European adoption
 - Industry recognition

1.7.4 5.4 IEEE

- **Standards:** IEEE 802.1X (Network Access Control), IEEE 1609 (WAVE Security)
- **Potential Collaboration:**
 - Participate in standards committees
 - Submit proposals for AI-enhanced security
- **Benefits:**
 - Academic and industry visibility

- Standards-compliant research
-

1.8 6. Open-Source Communities

1.8.1 6.1 ONOS Community

- **Project:** Open Network Operating System (SDN controller)
- **Potential Collaboration:**
 - Contribute AI-IDS app to ONOS
 - Present at ONOS Build events
 - Collaborate with SDN developers
- **Benefits:**
 - Community adoption
 - Real-world deployments
- **Repository:** <https://github.com/opennetworkinglab/onos>

1.8.2 6.2 OpenDaylight

- **Project:** Modular SDN controller
- **Potential Collaboration:**
 - Develop security service module
 - Join ODL community calls
- **Benefits:**
 - Alternative SDN platform
 - Diverse user base

1.8.3 6.3 TensorFlow/PyTorch Communities

- **Potential Collaboration:**
 - Contribute IDS models to TensorFlow Hub / PyTorch Hub
 - Write tutorials for network security ML
 - Participate in Google Summer of Code
- **Benefits:**
 - Showcase work to AI/ML community
 - Broader impact beyond security

1.8.4 6.4 Zeek (formerly Bro)

- **Project:** Open-source network security monitoring
 - **Potential Collaboration:**
 - Integrate AI models with Zeek
 - Develop Zeek plugin for ML-based detection
 - **Benefits:**
 - Large user base (enterprises, ISPs)
 - Established IDS platform
-

1.9 7. Data Sharing Agreements

1.9.1 7.1 Datasets Needed

- **5G Core Network Traffic:** Signaling messages, user plane data
- **IoT Botnet Traffic:** Real-world IoT malware propagation
- **Enterprise Network Logs:** Anonymized intrusion attempts
- **Industrial Control Systems (ICS):** SCADA/ICS attack traces

1.9.2 7.2 Data Sharing Partners

Telecom Operators: - Provide anonymized 5G traffic under NDA - Joint data analysis projects

CERTs (Computer Emergency Response Teams): - Access to incident response data - Collaborative threat intelligence

Canadian Institute for Cybersecurity (CIC): - CICIDS datasets provider - Collaborate on new dataset creation

1.9.3 7.3 Privacy and Legal Considerations

- **Anonymization:** Remove PII before sharing
 - **NDA:** Non-disclosure agreements with industry partners
 - **IRB Approval:** Obtain approval for human subjects data
 - **GDPR Compliance:** Follow EU data protection regulations
-

1.10 8. Collaboration Benefits

1.10.1 8.1 Benefits to Academic Research

- **Data Access:** Real-world datasets for validation
- **Equipment:** Hardware for testbed (saves \$10-20K)
- **Funding:** Sponsored research, fellowships
- **Deployment:** Real-world validation in production networks
- **Expertise:** Domain knowledge from practitioners
- **Career:** Industry connections for postdoc/employment

1.10.2 8.2 Benefits to Industry Partners

- **Innovation:** Early access to cutting-edge research
- **Talent Pipeline:** Recruit PhD students/postdocs
- **Product Enhancement:** Integrate research into products
- **Credibility:** Academic validation of technology
- **Publications:** Co-authored papers boost reputation
- **IP:** Potential patents from joint research

1.10.3 8.3 Benefits to Community

- **Open-Source:** Free, validated IDS for researchers and SMEs
 - **Knowledge:** Publications, tutorials, documentation
 - **Tools:** Reusable code, models, datasets
 - **Standards:** Influence on future security standards
-

1.11 9. Engagement Strategy

1.11.1 9.1 Timeline

Year 1: - Attend conferences, network with potential collaborators - Apply for cloud credits (AWS, Google, Microsoft) - Join open-source communities (ONOS, Zeek)

Year 2: - Reach out to industry partners for equipment donation - Apply for industry fellowships (Google, Microsoft, Cisco) - Propose collaborations with telecom operators

Year 3: - Deploy pilot IDS with industry partner - Engage with standards bodies (IETF, 3GPP) - Publish open-source framework

Year 4: - Finalize industry collaborations - Transfer technology to partners - Plan postdoc/industry position

1.11.2 9.2 Communication Channels

- **Email:** Direct outreach to researchers, product managers
- **LinkedIn:** Professional networking
- **Conferences:** Face-to-face meetings at IEEE, USENIX, ACM events
- **Workshops:** Present work at industry workshops
- **GitHub:** Open-source contributions
- **Blog/Medium:** Write about research for broader audience

1.11.3 9.3 Proposal Template

Email Template for Industry Collaboration:

Subject: PhD Research Collaboration: AI-Based IDS for 5G/IoT Networks

Dear [Name],

I am a PhD student at [University] working on AI-based intrusion detection for next-generation networks (5G/6G, IoT, SDN). My research aims to develop a distributed, privacy-preserving IDS using deep learning and federated learning.

I believe [Company] would be an excellent partner for this research. Potential collaboration areas include:

- [Specific area relevant to company]
- [Data sharing / Equipment / Funding]

- [Joint publications]

The benefits to [Company] would be:

- [Benefit 1: e.g., Enhanced security for products]
- [Benefit 2: e.g., Academic validation]
- [Benefit 3: e.g., Early access to research]

I would appreciate the opportunity to discuss this further. Are you available for a 15-minute call next week?

Best regards,

[Your Name]

PhD Candidate, [University]

[Email] | [Website] | [LinkedIn]

1.12 Conclusion

Successful PhD research requires strategic collaborations across academia, industry, standards bodies, and open-source communities. This document outlines a comprehensive engagement strategy to secure:

- **Data:** Real-world network traffic from telecom operators and enterprises
- **Equipment:** \$10-20K in donated hardware for testbed
- **Funding:** Fellowships, cloud credits, sponsored research
- **Validation:** Real-world deployment in production networks
- **Impact:** Industry adoption and standards influence

Key Partnerships to Pursue: 1. **Academic:** CMU, Berkeley, MIT for joint research 2. **Industry:** Cisco, Palo Alto Networks for equipment and funding 3. **Telecom:** Verizon, AT&T for data and deployment 4. **Cloud:** AWS, Google, Microsoft for computational resources 5. **Standards:** IETF, 3GPP for standardization 6. **Open-Source:** ONOS, Zeek for community adoption

Next Steps: - Year 1: Network at conferences, apply for cloud credits - Year 2: Reach out to industry for testbed equipment - Year 3: Deploy pilot IDS with partner, engage with standards - Year 4: Finalize collaborations, plan career next steps

Contact: idriss5214@github.com

Repository: github.com/idriss5214/AI-IDS-NextGen-Networks-PhD-Proposal

For more information, refer to: - ../proposal/full_proposal.md for research details - budget.md for financial planning - tools_and_frameworks.md for technical requirements